

Simon Aytes

Production AI Engineer with End-to-End Delivery Experience

New York, NY | (313) 806-6429 | simon@aytes.net | [/in/simonaytes](https://in/simonaytes) | saytes.io

PROFESSIONAL EXPERIENCE

SENIOR DATA SCIENTIST

January 2026 – Present

Penta Group, LLC — New York, NY (Hybrid)

- Owned architecture through production deployment of a full-stack AI tooling platform (FastAPI, React/TypeScript) scaling to 1,500+ monthly analysis runs across 180+ users company-wide; outputs feed nearly all client deliverables.
- Developed and shipped a production MCP server exposing the platform's analysis API to LLM agents, now powering live agentic workflows in client delivery.
- Deployed and maintained LLM-powered workflows for generation, summarization, document analysis, and analyst-facing automation; translated internal stakeholder requirements into reliable tools used in live delivery.
- Re-architected always-on EC2/SageMaker notebook pipelines into dynamically executed AWS ECS workers (PostgreSQL, CodePipeline, CodeBuild, CloudWatch), projected to cut infrastructure costs by 50%+.

AI ENGINEERING CONSULTANT

January 2025 – April 2026

Freelance — Remote

- Scoped, designed, and delivered a production LLM document-intelligence system end-to-end with full ownership, from discovery with non-technical stakeholders through production rollout.
- Built an invoice-processing system handling ~3,000 invoices/month across 100+ vendor formats, using LLM-driven OCR and OpenAI structured outputs with Pydantic schemas to convert unstructured documents into structured data.
- Delivered 2,000+ hours of manual-work savings and \$30,000+ in annual cost reduction; enabled the client team to scale without additional headcount.

GRADUATE RESEARCH ASSISTANT

February 2024 – December 2025

KAIST MLAI Lab — Seoul, South Korea

- First-authored Sketch-of-Thought: Efficient LLM Reasoning via Cognitive-Inspired Sketching, published at EMNLP 2025; introduced an inference-time reasoning method that reduced reasoning tokens by ~84% on average.
- Researched training-free inference-time methods and GRPO/SFT approaches for reducing LLM reasoning-token usage and improving test-time compute efficiency.

DATA SCIENTIST

September 2021 – February 2024

Penta Group, LLC — New York, NY

- Built NLP and social-listening systems for media intelligence, including automated topic extraction and large-scale text classification across social and news data; delivered client-facing analyses for Fortune 500 stakeholders.
- Embedded with internal teams to surface automation opportunities; built dashboards and reporting tools that reduced manual data-processing time by ~90%.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

December 2025

M.S., Artificial Intelligence, Advisor: Prof. Sung Ju Hwang

Lehman College, CUNY

December 2022

B.S., Computer Science (Minor: Data Science), Summa Cum Laude

TECHNICAL SKILLS

ML & AI

Python, PyTorch, LLMs, NLP, RAG, scikit-learn, Pandas, NumPy, SQL

TOOLS

OpenAI SDK, Anthropic SDK, FastAPI, PostgreSQL, TypeScript, React, Git

AGENTIC TOOLING

Claude Code, Codex, Cursor, MCP, Tool Use, Structured Outputs, Function Calling

INFRASTRUCTURE

AWS (ECS, EC2, S3, SageMaker, CodePipeline, CodeBuild), CI/CD, Docker, Linux